CodeAlpha E-Commerce Analysis

End-to-End Data Pipeline & Executive Dashboards

Project Overview & Architecture

Welcome to the **CodeAlpha E-Commerce Analysis** presentation. This project demonstrates a complete data engineering and analytics pipeline.

The Data Pipeline Stages:

- **Web Scraping:** Python & BeautifulSoup
- **Data Generation:** Relational synthetic data with Faker
- **Database:** SQLite & Normalized CSVs
- **EDA:** Exploratory Data Analysis & KPIs
- **Visualization:** Power BI Dashboards
- **Machine Learning:** NLP Sentiment Analysis

Phase 1: Data Engineering & Web Scraping

Simulated a realistic e-commerce environment by extracting live catalogs and augmenting with synthetic transactions.

Key Achievements:

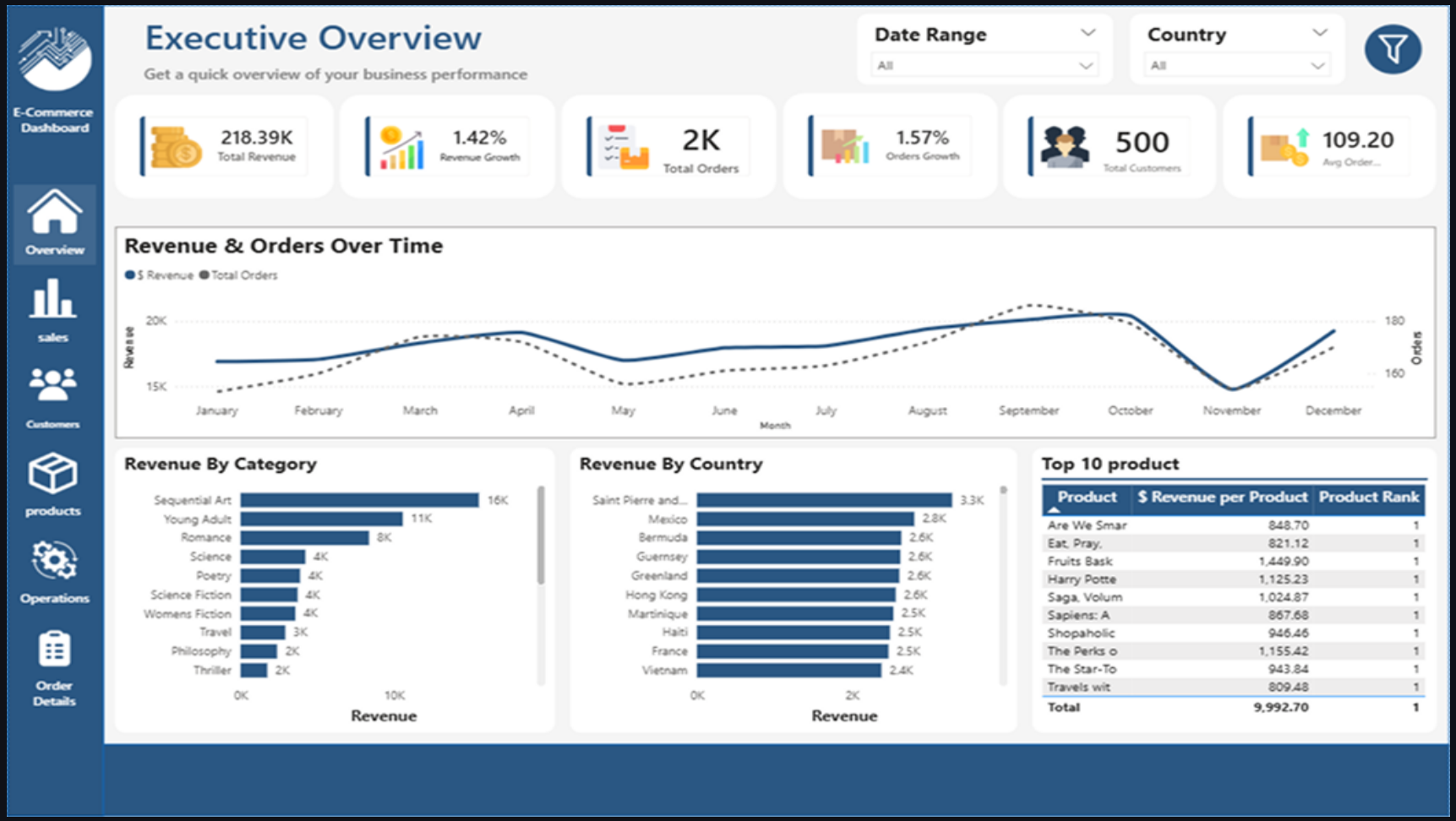
- **Extraction:** Scraped product details, prices, UPCs, and categories from books.toscrape.com.
- **Relational Data:** Generated 500+ customers, 2,000+ orders, and 4,000+ targeted NLP reviews.
- **Data Export:** Built a fully normalized SQLite Database with structured indexed tables.

Phase 2: Exploratory Data Analysis (EDA)

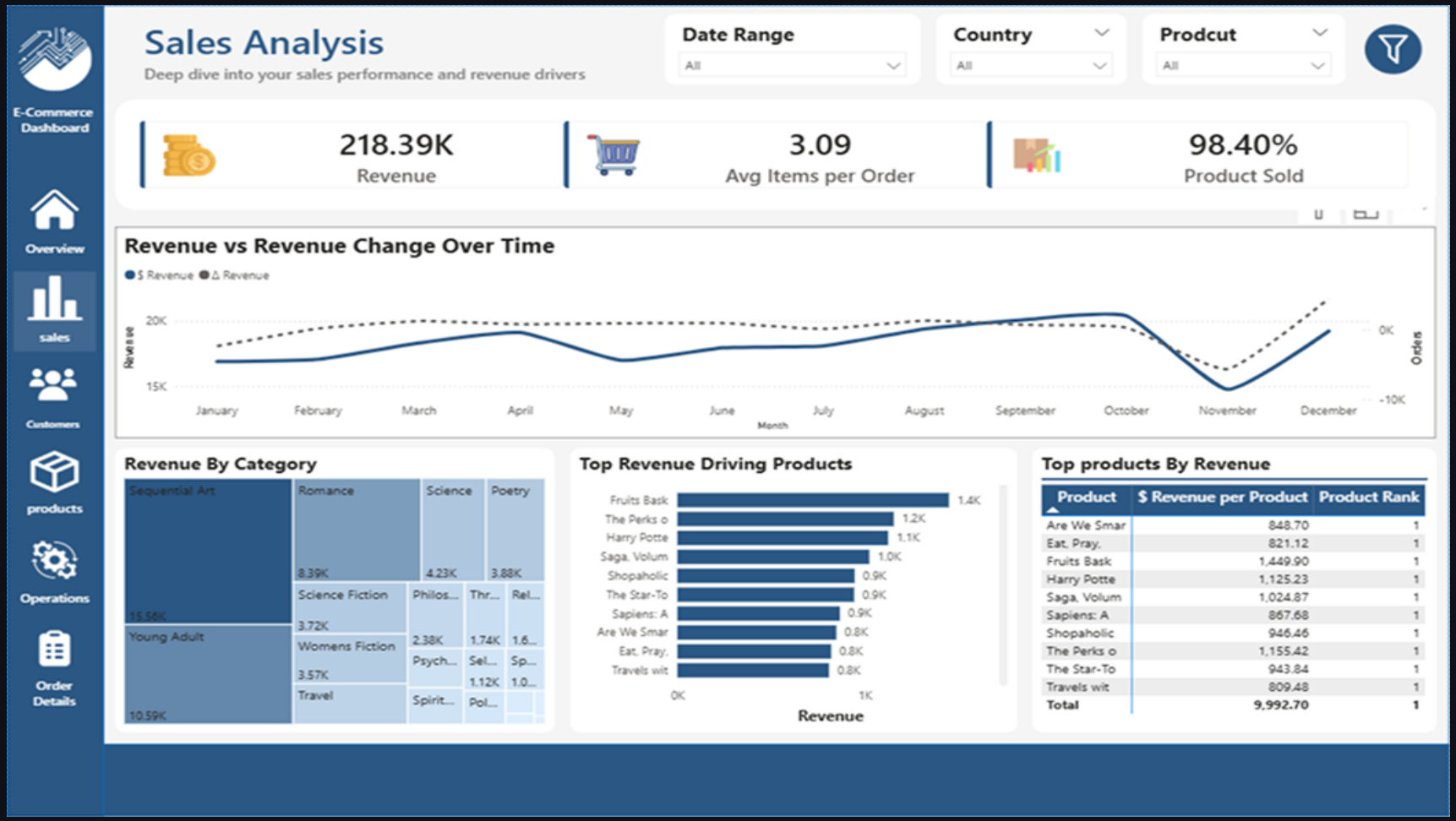
Core Business Insights:

- **The Pareto Principle:** A small fraction of products (the 20%) generates the vast majority of total revenue (the 80%).
- **Quality & Demand:** A highly positive correlation exists between high star ratings and overall sales volume.
- **Sales Seasonality:** Temporal peaks suggest strong seasonal purchasing behaviors.
- **Regional Dominance:** Specific geographic clusters drive the most consistent repeat purchases.

Phase 3: Executive Power BI Overview

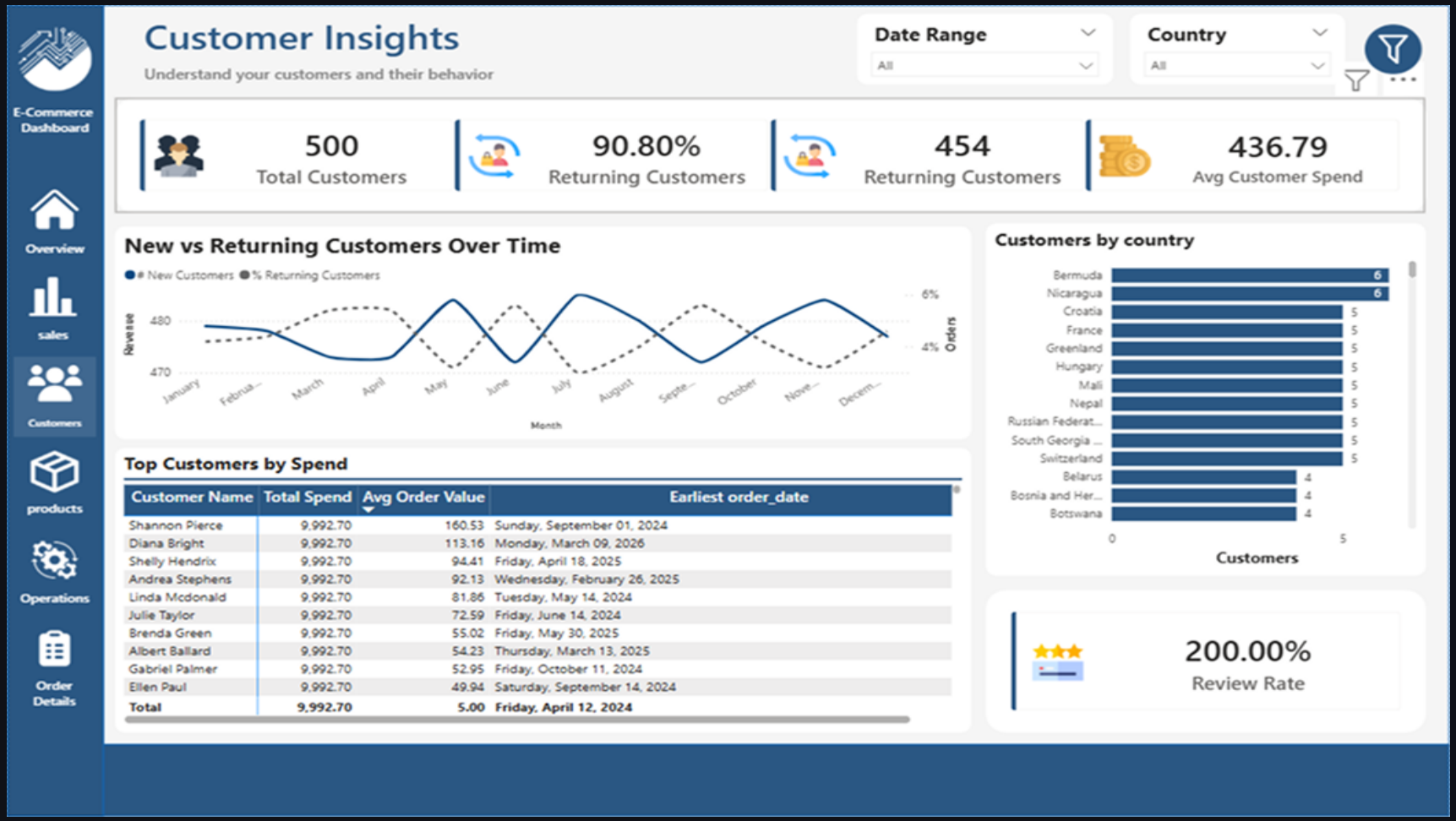


Dashboard: Sales Analytics

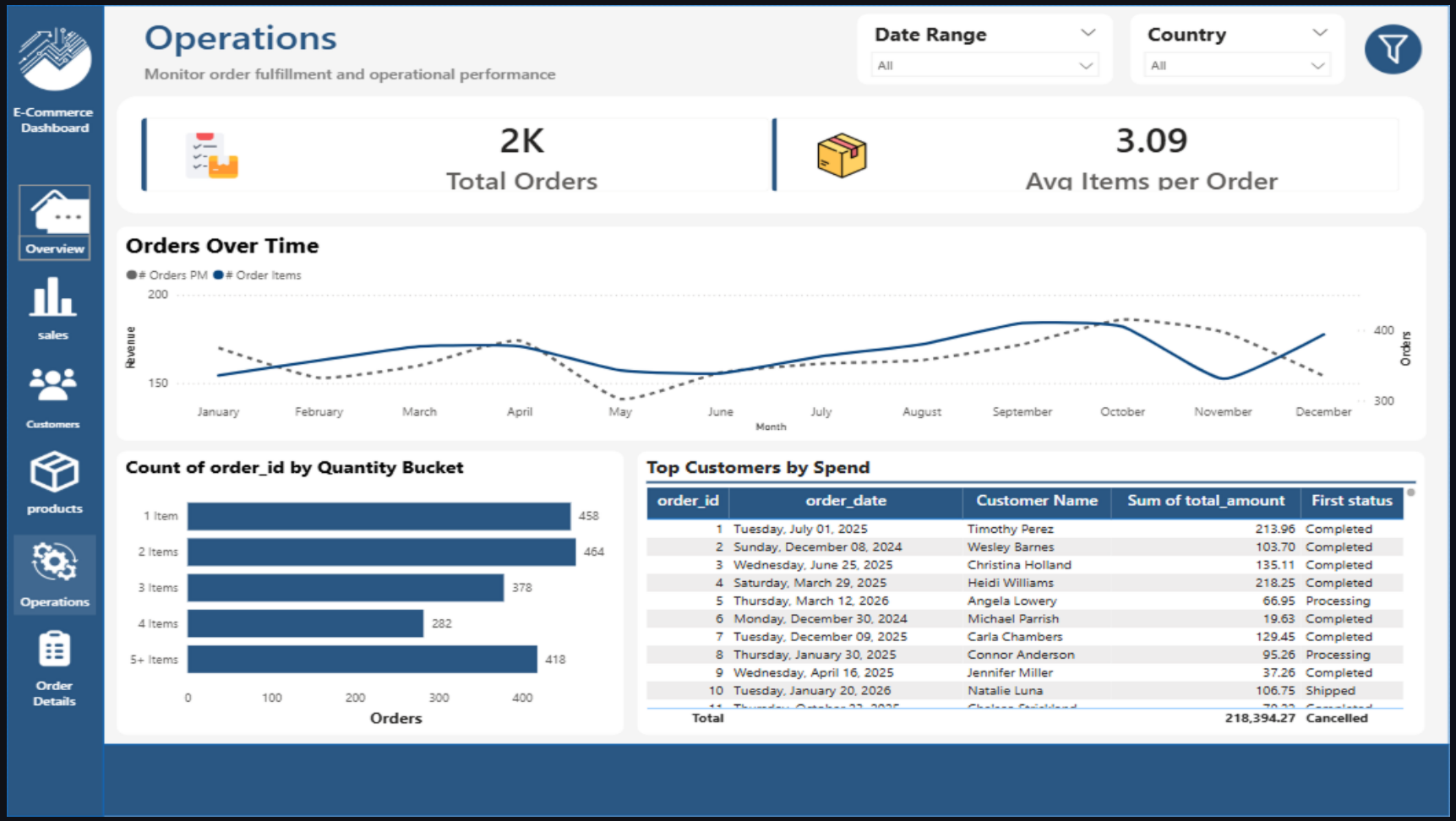


Dashboard: Product Analytics

Dashboard: Customer Demographics



Dashboard: Operations & Fulfillment



Phase 4: NLP Sentiment Analysis

Processed 4,000+ text reviews using Natural Language Processing (NLP) in Python.

1. Lexicon-Based Analysis:

- **VADER:** Extracted Compound Scores to tag reviews as Positive, Neutral, or Negative.
- **TextBlob:** Calculated Polarity and Subjectivity.
- **Emotion Detection:** Mapped text to Joy, Anger, Sadness, and Trust.

2. Machine Learning Classification:

- Deployed **TF-IDF Vectorization** coupled with a **Logistic Regression** classifier, achieving near-perfect sentiment prediction.

Conclusion & Strategic Impact

The integration of robust data engineering, clear visualizations, and predictive NLP provides a massive strategic advantage.

Business Recommendations:

1. **Targeted Marketing:** Leverage demographic data and positive sentiment drivers for ad targeting.
2. **Dynamic Inventory:** Use seasonal trends to ensure high-revenue products are in stock.
3. **Customer Retention:** Implement automated alerts for 'Negative' reviews to trigger interventions.